

SIMATS ENGINEERING

CO2-AT2-SCENARIO BASED ASSESSMENT

COURSE CODE: CSA0904

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COURSE NAME: Programming in JAVA

REG NO: 192421410

**SIMATS**
ENGINEERING

SIMATS Online Programming Lab
JAVA PROGRAMMING

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QUESTIONS

Q1
Student Registration System
10 marks

Q2
E-Commerce Cart System
10 marks

Q3
Library Management System
10 marks

Q4
Employee Data Processing
10 marks

Q5
Online Voting System
10 marks

5/5 answered

Q1 Student Registration System 10 marks

Suppose that a university system stores student data, where each student must be unique and can enroll in multiple courses.

(a) Enumerate three Collection interfaces suitable for this system.

(b) Design a Collection structure (diagram/representation) to store students and their enrolled courses.

(c) Write a Java program using HashSet and HashMap with Generics to:

- Add students
- Map students to courses
- Display data using Iterator

Your Code

```
1 import java.util.*;
2 public class StudentRegistration {
3     public static void main(String[] args) {
4         HashSet<String> students = new HashSet<String>();
5         students.add("Ramesh");
6         students.add("Suresh");
7         students.add("Priya");
8         students.add("Ramesh");
9         HashMap<String, String> courses = new HashMap<String, String>();
10        courses.put("Ramesh", "Java");
11        courses.put("Suresh", "Python");
12        courses.put("Priya", "Data Structures");
13        System.out.println("Students:");
14        Iterator<String> it = students.iterator();
```

Input

stdin input

OutputStudent Courses:
Suresh -> Python

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Q2 E-Commerce Cart System

10 marks

Suppose that an online shopping system maintains a cart where products are added, removed, and displayed in order.

- List three Collection classes applicable for cart management.
- Represent how cart items are stored using a List-based structure.
- Write a Java program using ArrayList and Iterator to:

- Add items
- Remove an item
- Display all items

Your Code

```
1 import java.util.*;
2 public class ShoppingCart {
3     public static void main(String[] args) {
4         ArrayList<String> cart = new ArrayList<String>();
5         cart.add("Laptop");
6         cart.add("Mobile");
7         cart.add("Headphones");
8         cart.add("Mouse");
9         System.out.println("Cart Items:");
10        Iterator<String> it = cart.iterator();
11        while (it.hasNext()) {
12            System.out.println(it.next());
13        }
14        cart.remove("Headphones");
15    }
16 }
```

Input

stdin input

Output

```
Cart Items:
Laptop
Mobile
Headphones
Mouse
```

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5/5 answered

Q3 Library Management System

10 marks

Suppose that a library stores books with unique ISBN numbers and needs fast retrieval.

- Name three Map implementations in Java.
- Design a schema-like representation using Map for storing ISBN and book details.
- Write a Java program using HashMap to:

- Insert book details
- Retrieve a book
- Display all books

Your Code

```
1 import java.util.*;
2 public class LibraryManagement {
3     public static void main(String[] args) {
4         HashMap<String, String> books = new HashMap<String, String>();
5         books.put("ISBN101", "Java Programming");
6         books.put("ISBN102", "Data Structures");
7         books.put("ISBN103", "Computer Networks");
8         System.out.println("Book Retrieved:");
9         System.out.println("ISBN102 -> " + books.get("ISBN102"));
10        System.out.println("\nAll Books:");
11        for (Map.Entry<String, String> entry : books.entrySet()) {
12            System.out.println(entry.getKey() + " -> " + entry.getValue());
13        }
14    }
15 }
```

Input

stdin input

Output

```
Book Retrieved:
ISBN102 -> Data Structures

All Books:
ISBN103 -> Computer Networks
ISBN102 -> Data Structures
```

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10 marks

5/5 answered

Q4 Employee Data Processing

10 marks

Suppose that a company processes employee records and filters employees based on salary.

- List three advantages of Generics in Collections.
- Represent how employee data is stored using List with Generics.
- Write a Java program using Iterator to:
 - Store employee objects
 - Display employees with salary > 50,000

Your Code

```
1 import java.util.*;
2 public class Employee {
3     public static void main(String[] args) {
4         List<String> names = new ArrayList<String>();
5         List<Double> salaries = new ArrayList<Double>();
6         names.add("Ramesh");
7         salaries.add(45000.0);
8         names.add("Suresh");
9         salaries.add(60000.0);
10        names.add("Priya");
11        salaries.add(55000.0);
12        names.add("Anu");
13        salaries.add(40000.0);
14        System.out.println("All Employees:");
15        Iterator<String> it = names.iterator();
```

Input

stdin input

Output

```
All Employees:
Ramesh - 45000.0
Suresh - 60000.0
Priya - 55000.0
Anu - 40000.0
```

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5/5 answered

Q5 Online Voting System

10 marks

Suppose that an online voting system ensures one vote per user and counts votes efficiently.

- Identify three suitable Collection types for this system.
- Design a structure using Set and Map for voters and vote counting.
- Write a Java program using HashSet and HashMap to:
 - Record votes
 - Prevent duplicate voting
 - Display results

Your Code

```
1 import java.util.*;
2 public class OnlineVoting {
3     public static void main(String[] args) {
4         HashSet<String> voters = new HashSet<String>();
5         HashMap<String, Integer> votes = new HashMap<String, Integer>();
6         votes.put("Candidate A", 0);
7         votes.put("Candidate B", 0);
8         votes.put("Candidate C", 0);
9         String[] voterNames = {
10             "Ramesh", "Suresh", "Priya", "Anu", "Ramesh"
11         };
12         String[] choices = {
13             "Candidate A", "Candidate B", "Candidate A",
14             "Candidate C", "Candidate B"
15         };
```

Input

stdin input

Output

```
Ramesh voted for Candidate A
Suresh voted for Candidate B
Priya voted for Candidate A
Anu voted for Candidate C
Ramesh - Duplicate vote rejected
```